

Tidy a Bookshelf SOP (1x Shelf, in situ)

One episode tidies one shelf of a real bookcase, in place.

The base is **passive**. It has no drive of its own. It is pushed by hand to the bookcase and locked there. Nothing is carried away to a table. Everything the episode touches is already on the bookcase when recording starts.

The shelf is worked **as found**. Six loose books lie flat in stacks on the shelf board. Three non-books sit among them. Dust runs along the shelf lip. The episode ends with all six books standing upright in one row, tallest to shortest, spines facing out, held up by a bookend. The three non-books are in the bin. The lip is clean.

This is an in-situ task. Two rules follow from that.

First, there is a shelf directly above the one being worked. So every approach is from the front, straight in through the shelf opening. No gripper ever comes down onto the shelf board from above.

Second, no book is ever lifted clear of the shelf. Books are pushed along the board, tipped up on the board, and pressed into the row on the board. Only the non-books, the cloth, and the bookend leave the shelf board. None of those is a book.

The order never changes. Clear the non-books into the bin. Dust the shelf lip. Shelve the six books, tallest first. Set the bookend last. The shelf is cleared before it is dusted. It is dusted before any book is stood up on it.

The two arms never cross. The left arm works the left. The right arm works the right. The left gripper always stays left of the right gripper.

Arm roles are fixed and never swap. The **right gripper is the book gripper**. It tips every book up. The **left gripper is the supporting gripper**. It clears the non-books into the bin. It feeds each book along the board. It stands as the **stop** the book pivots against. It presses the standing book into the row. It sets the bookend.

The row is built from the **right end panel leftward**, tallest book first. Every new book joins on the open left side. No book is ever worked over a book already standing. The bookend goes on last.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.

- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the bookcase. Stop it **square to the front of the bookcase**, so the shelf lip runs straight across the frame and neither end of the shelf sits nearer than the other.
2. Stop it **centered on the working shelf**, so the right end panel and the left end of the working span are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the shelf back straight in through the shelf opening without either arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the bookcase when both arms work at once.
4. Check the **working shelf height**. With the base parked, both grippers must enter the shelf opening level, reach the shelf board and the shelf back, and travel the whole working span without the wrist or forearm touching the overhead shelf or the shelf lip.
5. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
6. If any of lines 1 to 4 fails, push the base to a new park by hand and start again at line 1. Do not work a shelf the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the furniture hard enough to shift it. No operator touches it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the working shelf. Its frame includes the whole shelf board from the left end of the working span to the right end panel, the whole shelf lip, the shelf opening, and the bin, the bookend spot, and the cloth rest on the lower shelf.
3. The camera reads the shelf lip from the front. You can see the dust on it and whether it has been wiped off.
4. The camera reads each book from the front. You can see its spine, whether it lies flat or stands, whether it is the right way up, and how long it is against its neighbours.
5. Both arms are at home with grippers open.
6. The bookcase is steady. It does not rock when a book is pushed along the shelf board.
7. The right arm reaches the build place, the strip of shelf board from the build place to the right end panel, and the shelf back along that strip. It reaches all of them straight in from the

front and without extending to a joint limit. It reaches nothing on the lower shelf but the cloth rest.

8. The left arm reaches the three non-books, every stack on the working shelf, the build place, the left cover of every book in the finished row, and the bin, the bookend spot, and the cloth rest on the lower shelf. It reaches all of them without extending to a joint limit. It never needs to reach the right end panel.
9. Each arm enters and leaves the shelf opening **from the front and level**. Neither arm can come down onto the shelf board from above. Neither wrist nor forearm touches the overhead shelf or the shelf lip on the way in or out.
10. Both arms work the build place at the same time without touching each other or the bookcase, with the left gripper on the left and the right gripper on the right.
11. If a zone cannot be reached, re-park the base by the Base positioning steps until lines 7 to 10 hold.

Materials checklist

1. The **working shelf** is one shelf of the bookcase with a **shelf directly above it**. So the way in is the front-facing **shelf opening** and never the top.
2. The shelf opening is tall enough that the tallest book stands on the board with clear space above it. It is tall enough that a book tipping up from flat sweeps through without touching the overhead shelf.
3. The shelf board is level and bare. It carries **visible dust along the shelf lip** from end to end. The dust reads on camera from the front.
4. The shelf board carries no labels, marks, tape, dividers, or slots. You judge every place on it by eye against the right end panel, the shelf lip, and the books already standing.
5. The **right end panel** is fixed. It stands taller than the tallest book. The row is pressed against it. It is the only thing the row leans on until the bookend goes in.
6. The working shelf is long enough to hold the three stacks, the clear board at the build place, and the finished row plus bookend all at once, with clear board still left over at the left end.
7. **Six loose books** lie on the shelf board in **three stacks of two**. The stacks are spread along the left part of the shelf, clear of the build place and clear of the right end panel. No stack touches another.
8. Every loose book **lies flat on a cover**. Its **spine points toward the front of the shelf**. Its **bottom edge is to the left**. Tipping it up to the left then stands it the right way up. No loose book stands, leans, lies spine up, or lies with its spine toward the back.
9. **Each stack has the taller of its two books on top**. This holds for all three stacks, with no exception. So the tallest book on the shelf is always one that is already lying free on top.
10. The **six books are all visibly different in height**. Each is clearly longer than the next as they lie, so which one is longest is never a close call. Their lengths read on camera from the front.

11. No loose book overhangs the shelf lip. No loose book is pushed hard against the shelf back.
12. Every book stands on its own bottom edge unaided on a level surface. It does not sag or flop when held by its covers. It is thick enough for a gripper to close on its two covers without crushing it.
13. Every book is short enough front to back to stand fully on the shelf board with its page edge clear of the shelf back.
14. **Three non-books** sit on the shelf board among the stacks. All three are **left of the shelf midline**. They are one mug, one small picture frame, and one coiled cable. No two touch. Each has a gripper's width of clear space beside it. Nothing else on the shelf is a non-book.
15. Each non-book sits steady on the bare shelf board on its own. It is closed and dry. One gripper lifts it without its contents spilling or its parts coming apart.
16. The **lower shelf** is the shelf directly below the working shelf. It holds three things and nothing else. The **bin** sits at its left end. The **bookend spot** sits next along, between the bin and the middle. The **cloth rest** sits at its middle. The right end of the lower shelf is bare. Nothing on it is marked or taped out.
17. The bin stands open at the left end of the lower shelf. Its mouth is clear and faces up. It is empty. No non-book begins inside it. It is deep enough for all three non-books. It does not tip when one is released into it.
18. The bookend spot sits between the bin and the cloth rest, where the **left gripper** reaches it without extending. It holds the bookend and nothing else.
19. The folded cloth lies flat and damp on the cloth rest at the middle of the lower shelf. **Both** grippers reach the rest without extending. The rest holds nothing else.
20. The **bookend** is a **solid block** with two flat parallel faces and its own flat base. It has no foot that slides under the books. It stands upright unaided. It does not rock. It is heavy enough to hold six books up on its own.
21. The bookend stands upright on its own base on the bookend spot, with its flat faces facing left and right. That is the same way it will stand on the working shelf, so it never has to be turned.
22. The lower shelf has clear space above the bin, the bookend spot, and the cloth rest. The left gripper reaches into the bin and onto the bookend spot straight in from the front. Both grippers reach the cloth rest straight in from the front.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the bookcase itself: the shelf lip, the right end panel, and the objects already in place.

- **Bookcase:** the unit the base is parked at. It is never moved, never leaned on, and never pushed.
- **Working shelf:** the shelf being tidied. All the book work happens on it.

- **Overhead shelf:** the shelf directly above the working shelf. It is what makes every approach a front approach. Nothing ever touches it.
- **Shelf opening:** the front-facing gap between the working shelf board and the overhead shelf. Every arm goes in and comes out through it, level and from the front.
- **Shelf board:** the flat surface of the working shelf. Books are pushed along it, tipped up on it, and pressed into the row on it. No book leaves it.
- **Shelf lip:** the narrow front edge face of the working shelf board. It runs the full length and faces the front. It is the only surface that is dusted.
- **Shelf back:** the panel behind the working shelf. Books stand clear of it.
- **Right end panel:** the fixed side of the bookcase at the right end of the working shelf. The row is pressed against it. The first book stands against it.
- **Shelf midline:** the point halfway along the working shelf, judged by eye. It splits the shelf lip into a **left half** and a **right half**, one for each gripper.
- **Row:** the books standing on the shelf board. It starts at the right end panel and grows leftward. The tallest book stands at the panel. The shortest stands at the left end.
- **Build place:** the stretch of clear shelf board immediately left of the row, at the front of the board. Every book is fed here, tipped up here, and pressed from here into the row. It moves left as the row grows.
- **Stacks:** the three piles of two flat books, spread along the left part of the shelf board, clear of the build place. They are gone by the end of the episode.
- **Lower shelf:** the shelf directly below the working shelf. It holds the **bin** at the left end, the **bookend spot** next along, and the **cloth rest** at the middle. Its right end is bare.
- **Bin:** the open bin at the left end of the lower shelf. Empty at episode start. Holds the three non-books at episode end. **Left gripper only.**
- **Bookend spot:** the place between the bin and the cloth rest where the bookend stands at episode start. Empty at episode end. **Left gripper only.**
- **Cloth rest:** the place at the middle of the lower shelf where the folded cloth lies at episode start and at episode end. Both grippers reach it. It is the only place either gripper sets the cloth down.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The left arm works the left. The right arm works the right.
- The **left gripper always stays left of the right gripper**. When both are on the shelf at once, the left gripper is nearer the left end and the right gripper is nearer the right end.
- Neither arm reaches over, under, around, or past the other.
- Neither arm reaches across the front of the other arm's body.

- Only the cloth passes between the grippers. It passes only by being laid on the cloth rest and taken up again. Nothing is ever handed from gripper to gripper in the air.

Arm assignments

- **Right gripper. The book gripper.** It tips every book up at the build place. It dusts the right half of the shelf lip. It presses each standing book back off the lip. It never touches the bin, the bookend spot, a stack, a non-book, or the bookend. On the lower shelf it touches only the cloth rest.
- **Left gripper. The supporting gripper.** It clears the three non-books into the bin. It dusts the left half of the shelf lip. It feeds each book along the board to the build place. It stands as the **stop** for the tip-up. It presses each standing book into the row. It brings in the bookend and pushes it home. It never touches the right end panel.
- Only one arm moves an object at a time.

Vocabulary

- **Book:** a bound stack of pages with a front cover, a back cover, and a spine. Anything on the working shelf that is not a book is a **non-book**.
- **Non-book (observable list):** for this task the non-books are exactly three. They are one mug, one small picture frame, and one coiled cable. Nothing else on the shelf is a non-book. No book is ever treated as one. This list removes judgment calls and is **unvalidated**.
- **Spine:** the bound back of the book, the narrow face that carries the title. It is the face that must end up looking out the front of the shelf.
- **Page edge:** the open edge of the book, opposite the spine.
- **Bottom edge:** the edge a book stands on when it is shelved. **Top edge** is the edge opposite it. On a flat book lying spine forward, these are the short edges at its left and right ends.
- **Covers:** the two large flat faces of the book. A gripper closes on these two faces. It never closes on the pages, the spine, the page edge, or a corner.
- **Free book:** a book with nothing lying on top of it. It is either the top book of a stack or a book lying alone on the shelf board. Only free books are worked. A book with another book on it is never dug out.
- **Longest free book:** the free book that measures longest left to right as it lies. A flat book's left to right length is its height when it stands. So the longest free book is the tallest one available. The six lengths are visibly different, so there is never a tie.
- **Overhang the lip:** the book's spine edge sticks out past the shelf lip far enough that a gripper finger passes under it in free air. This is what makes the book grippable. A book lying flat on the board has no space beneath it.

- **The stop:** the **left gripper**, closed, standing on the shelf board hard against the left end of a flat book, so that end cannot slide left. It makes the book pivot instead of skate. It is a block, not a grip. The gripper never closes on a book.
- **Tip-up:** standing a flat book on its bottom edge. The right gripper lifts the book's top edge and swings it over the bottom edge. The bottom edge stays on the board against the stop. The book does not leave the board. It is not turned any other way. The tip-up ends with the book **leaning** on the stop, not standing straight.
- **Steadying press:** the **left gripper**, closed, held against the left cover of the leftmost book with light pressure toward the right end panel. It brings the book upright and stops the row falling to the left. It is a press, never a grip.
- **Front approach:** the gripper enters and leaves the shelf opening level and from the front. It never comes down onto the shelf board from above. Every action on the working shelf is a front approach.
- **Spine out:** the book stands on its bottom edge with its spine facing the front of the shelf, so the title reads from the front. A book showing a cover at the front, or showing its page edge, is not spine out.
- **Right way up:** the book stands with its spine title reading the way it would on any shelf, not upside down.
- **Height order:** each book in the row is shorter than the one to its right. The tallest stands against the right end panel. The shortest stands at the left end of the row.
- **Snug:** every book in the row touches its neighbour cover to cover. No gap shows between any two books. No gap shows between the tallest book and the right end panel.
- **Set back off the lip:** the book's spine sits just inside the shelf lip. It does not stick out past it.
- **Stands on its own:** after the grippers move clear, the book stays upright for 2 seconds. It does not rock, lean further, slide, or fall.
- **Dusted:** no dust is visible anywhere along the shelf lip when it is read from the front, end to end.

Steps

Step 1: Clear the non-books into the bin

Goal: the shelf board holds nothing but the three stacks of books. All three non-books are in the bin.

All three non-books are left of the shelf midline. The **left gripper** does this whole step. The right arm stays at home.

- Work one non-book at a time. Never lift two together.
- Take the non-book nearest the shelf lip first, then the next nearest, then the last.

- With the **left gripper**, enter the shelf opening from the front. Close on the non-book at its widest visible point. Close only hard enough to hold it without dropping it.
- Draw it straight out of the shelf opening toward the front, level. Do not lift it toward the overhead shelf. Do not drag it over a stack.
- Carry it down to the bin at the left end of the lower shelf. Do not carry it over the bookend spot or the cloth rest.
- Hold it over the bin mouth. Open the **left gripper** so it drops in. Do not throw it. Do not let go outside the bin mouth.
- If a non-book misses the bin, pick it up with the **left gripper** and bin it again, once.
- Repeat until no mug, picture frame, or coiled cable is left on the shelf board.

Check: the **left gripper** is clear of the shelf. The shelf board holds nothing but the three stacks. All three non-books are in the bin. If a non-book is still on the board, bin it now before Step 2.

Expected state: three stacks of two flat books, dust still on the shelf lip, three non-books in the bin.

Step 2: Dust the shelf lip

Goal: no dust is visible anywhere along the shelf lip. The cloth is back on its rest.

The lip is on the outside of the bookcase. So this step is worked in front of the shelf opening, not inside it. Each gripper dusts the half of the lip on its own side of the shelf midline. Neither gripper crosses the midline. The cloth is never handed across. It goes back on the rest and is taken up again.

2.1 Dust the left half of the lip

- With the **left gripper**, close on the folded cloth on the cloth rest and lift it clear.
- Keep it folded and keep it flat. Do not let it unfold, trail, or catch on a shelf edge.
- Press the cloth flat against the shelf lip at the shelf midline.
- Drag it along the lip to the left, all the way to the left end of the shelf. Keep it pressed on the lip the whole way.
- Drag it back to the shelf midline the same way.
- Do that pass twice. Keep the cloth on the lip face only. Do not run it along the shelf board. Do not push it into the shelf opening. Do not lean on the bookcase.
- Lay the cloth flat back on the cloth rest. Open the **left gripper** and move clear.

Check: the **left gripper** is clear. No dust is visible on the left half of the lip. If dust remains, take the cloth off the rest with the **left gripper** and repeat this substep once.

2.2 Dust the right half of the lip

- With the **right gripper**, close on the folded cloth on the cloth rest and lift it clear.

- Press the cloth flat against the shelf lip at the shelf midline.
- Drag it along the lip to the right, all the way to the right end of the shelf. Keep it pressed on the lip the whole way.
- Drag it back to the shelf midline the same way.
- Do that pass twice, under the same rules as Step 2.1.
- Lay the cloth flat back on the cloth rest. Open the **right gripper** and move clear.

Check: the shelf lip reads clean from end to end. The cloth lies flat and folded on its rest.

Expected state: clean lip, three stacks of two flat books on the board, bookend still on its spot.

Step 3: Shelve the six books, tallest first

Goal: all six books stand upright in one row on the shelf board. All are spine out, right way up, and in height order. The tallest is snug against the right end panel.

Run Steps 3.1 to 3.4 once per book, six times. **No book leaves the shelf board at any point.**

Through this whole step the **left gripper stays left of the right gripper**. The left gripper works the book's left end. The right gripper works the book's right end. Neither reaches past the other.

3.1 Choose the longest free book

- Read the six books as they lie. Pick out the **longest free book**. That is the longest one with nothing on top of it.
- Every stack has its taller book on top. So the longest free book is always the tallest book still to be shelved.
- **Never move a book to get at the one under it.** The book underneath becomes free on its own once the book above it has gone.

3.2 Feed the book to the build place

- With the **left gripper** closed, enter the shelf opening from the front. Set the gripper against the **left end** of that book.
- Push the book to the right along the board. If it was the top book of a stack, keep pushing until it slides off the book beneath it and settles flat on the shelf board. Then keep going.
- Push it as far as the **build place**, the clear board just left of the row. Do not push it into the row. Do not push it against the right end panel.
- Move the **left gripper** round to the book's back edge. Push the book **forward** until its spine edge **overhangs the shelf lip**. Push it far enough that a gripper finger passes under it, and no further.
- Keep the book flat on the board the whole time. Never lift it.

Check: the book lies flat at the build place. Its spine overhangs the lip. Its spine still points to the front and its bottom edge is still to the left. If it landed turned, square it with the closed **left**

gripper against its edges. If its spine ended up toward the back, or it landed on the wrong cover, push it back off the lip with the **left gripper** and leave it for the station. Do not tip up a book that will stand wrong.

3.3 Tip the book up against the stop

- Bring the **left gripper**, closed, against the **left end** of the book. Hold it there as **the stop**. It does not move again until the book is standing.
- With the **right gripper**, come in from the front. Close on the book's two covers on the **overhanging spine strip, near its right end**. The lower finger passes under the book in the free air past the lip.
- The right gripper works only the right end of the book. It never reaches past the left gripper.
- Do not close on the pages, the page edge, or a corner. Do not squeeze hard enough to bend the covers.
- Lift that end up and swing it **to the left**, over the stop. The book pivots on its bottom edge, which stays on the board against the **left gripper**.
- Keep the pivot edge on the board. The book never leaves the shelf board. It is never turned any other way.
- Stop as soon as the book **leans** against the **left gripper**. Do not carry it all the way upright with the right gripper.
- Open the **right gripper** and draw it straight out of the shelf opening, clear of the book.

Check: the book leans on the **left gripper**. It stands on its bottom edge, spine toward the front, right way up. If it slid instead of pivoting, put the **left gripper** back hard against its left end and repeat this substep once.

3.4 Press the book into the row and set it back off the lip

- Keep the **left gripper** on the book's left cover. Press it to the right until the book comes upright and meets what is beside it. For the first book that is the **right end panel**. For every book after it, that is the last book placed.
- Press until the row is snug against the right end panel, with no gap between any two books. Stop as soon as the books stop moving.
- With the **right gripper** closed, come in from the front. Press the new book's spine straight back until it sits just inside the shelf lip, level with the spines beside it. Press on the right of the spine face, right of where the left gripper is holding.
- Draw the **right gripper** straight out of the shelf opening.
- Lift the **left gripper** off the book and draw it straight out. The row leans right against the right end panel. It stands on its own while the left gripper is away.

Check after each book: the book stands on its bottom edge, spine out and right way up. Its top edge is lower than the top edge of the book to its right. No gap shows beside it. No gap shows

at the right end panel. No spine sticks out past the lip.

If it leans or a gap shows, press again with the **left gripper**. If it stands spine in, upside down, or out of height order, hold the row with the **left gripper**, lay the book back down flat with the **right gripper**, and run Steps 3.2 to 3.4 again for it.

Expected state after the sixth book: six books stand in one row from the right end panel leftward. Their top edges step down right to left. All spines face out and read the right way up. All spines sit level just inside the lip. No book is left flat on the shelf board.

Step 4: Set the bookend

Goal: the bookend stands on the shelf board at the left end of the row. Its right face is against the leftmost book. It holds all six books upright and snug against the right end panel.

The bookend goes on the left end of the row. So the **left gripper** does this whole step. The right arm stays clear of the shelf.

4.1 Bring the bookend in

- With the **left gripper**, close on the bookend's two flat faces on the bookend spot. Lift it clear of the lower shelf.
- Keep it upright on its own base with its flat faces facing left and right. Do not turn it.
- Carry it up in front of the shelf opening. Take it in from the front, level. Do not pass it over a book.
- Set it down on the shelf board just to the left of the leftmost book. Stand it on its own base, close beside the book but not touching it.
- Open the **left gripper** and draw it straight out of the shelf opening.

4.2 Push the row snug

- Bring the **left gripper**, closed, in from the front. Set it against the bookend's **left face, low on the face**, so the push does not tip the block.
- Push the bookend to the right along the board until its right face meets the leftmost book's cover and the whole row stands upright and snug against the right end panel.
- Push only until the books stop moving. Do not keep pushing. Do not push hard enough to move the bookcase.
- Draw the **left gripper** straight out of the shelf opening.

Check: the bookend stands on its own base with its right face against the leftmost book. All six books stand upright with none leaning. No gap shows between any two books. No gap shows at the right end panel.

If a book still leans or a gap shows, bring the **left gripper** back low against the bookend's left face and push once more. If a book is spine in, upside down, or out of height order, hold the row

with the **left gripper**, lay that book back down flat with the **right gripper**, and run Steps 3.2 to 3.4 again for it.

Expected state: clean lip. Six books in one row, tallest at the right end panel and shortest at the bookend. All spines out and level just inside the lip. Bookend at the left end. Three non-books in the bin. Cloth on its rest. Bookend spot empty. Clear bare board at the left of the shelf.

Step 5: End the episode

- Confirm the shelf board. It holds nothing but the six books and the bookend. No book is left lying flat.
- Confirm the row. All six books stand on their bottom edges, in one row, each spine out and each book the right way up.
- Confirm the order. Each book's top edge is lower than the top edge of the book to its right. The tallest stands against the right end panel.
- Confirm the row is snug. No gap between any two books. No gap at the right end panel. The bookend's right face is against the leftmost book.
- Confirm the fronts. Every spine sits just inside the shelf lip. None sticks out past it.
- Confirm the lip. It reads clean from end to end.
- Confirm the lower shelf. All three non-books are in the bin. The cloth lies flat on its rest. The bookend spot is empty.
- Confirm the base has not moved and the bookcase has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the bookend off the shelf board. Stand it upright on the bookend spot, on its own base, flat faces facing left and right.
2. Take the six books off the row. Lay them back on the shelf board as **three stacks of two**, spread along the left part of the shelf, clear of the build place and clear of the right end panel. No stack touches another.
3. Lay every book flat on a cover, with its **spine toward the front** and its **bottom edge to the left**. No book overhangs the shelf lip.
4. Put the **taller book of each pair on top**. Check the three stacks between them use all six books.
5. Take the three non-books out of the bin. Set them on the shelf board among the stacks, all of them left of the shelf midline. None touches another. Each has a gripper's width of clear space beside it. Leave the bin empty.

6. Put dust back along the shelf lip from end to end, so the next episode has something to wipe off. Check it reads on camera from the front.
7. Rinse or replace the cloth. Fold it flat and damp and lay it on the cloth rest.
8. Replace a book that will no longer stand on its own bottom edge unaided, that sags or flops when held by its covers, or whose covers a correct grip will no longer hold.
9. Replace a bookend that rocks on its base, is too light to hold six books, or slides along the board under a correct low push instead of moving the row.
10. Replace a non-book that comes apart or will no longer sit steady on the bare shelf board.
11. Clear any debris off the shelf board and the lower shelf. Check the bookcase is still steady and the right end panel still fixed.
12. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the bookcase shifts in frame, the shelf lip changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 5. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the left gripper goes to the right of the right gripper, or the right gripper goes to the left of the left gripper. One arm reaches over, under, around, or past the other. One arm

reaches across the front of the other arm's body. The right arm goes to the bin, the bookend spot, or a stack. The left arm reaches the right end panel.

- **SOP rule broken:** Arm lanes and Arm assignments. The left arm works the left, the right arm works the right, and the left gripper always stays left of the right gripper.
- **Coaching note:** each arm stays in its own lane. If you have to reach past the other arm, the wrong arm is doing the job.

Violation: Approached from above or touched the overhead shelf

- **Visible cue:** a gripper comes down onto the shelf board from above instead of entering level from the front. Or a wrist or forearm knocks, scrapes, or leans on the overhead shelf or the shelf lip on the way in or out.
- **SOP rule broken:** Steps 1 to 4. Every action on the working shelf is a front approach, level, in and out through the shelf opening.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Book lifted off the shelf board

- **Visible cue:** a book leaves the shelf board. It is carried out of the shelf opening, held clear of the board in mid-air, or set down somewhere other than where it was pushed.
- **SOP rule broken:** Steps 3.2 to 3.4. Books are pushed, tipped, and pressed on the board and never lifted clear of it.
- **Coaching note:** the board carries the book the whole way. Only the non-books, the cloth, and the bookend leave it.

Violation: Wrong order

- **Visible cue:** the lip is dusted before the non-books are cleared. A book is stood up before the lip is dusted. The bookend goes in before the sixth book is standing. Or a step is skipped.
- **SOP rule broken:** Steps 1 to 4. Clear the non-books, dust the lip, shelve the six books, then set the bookend.
- **Coaching note:** clear, dust, shelve, bookend. Nothing stands on a shelf that has not been dusted.

Violation: Non-book left on the shelf or a book binned

- **Visible cue:** the mug, the picture frame, or the coiled cable is still on the shelf board when the **left gripper** moves on to the cloth or to a stack. Or a book is carried to the bin.
- **SOP rule broken:** Step 1. Clear all three non-books off the shelf board into the bin and bin nothing else.

- **Coaching note:** the three non-books go in the bin and nothing else does. Read the whole board again before moving on.

Violation: Non-book thrown or released outside the bin

- **Visible cue:** the **left gripper** tosses a non-book. It lets go of it away from the bin mouth so it lands on the lower shelf or the floor. It drags it over a stack. It carries it over the bookend spot or the cloth rest. Or it leaves it where it landed after a miss.
- **SOP rule broken:** Step 1. Draw each non-book out level, carry it clear of the other places, hold it over the bin mouth, let go into the bin, and pick up a miss once.
- **Coaching note:** over the mouth before letting go, and always pick up a miss.

Violation: Cloth handled or set down wrong

- **Visible cue:** the cloth unfolds, trails, or catches on a shelf edge. It is dropped. It is handed from one gripper to the other instead of being laid on the cloth rest. It is set down anywhere other than the rest. Or it is pushed into the shelf opening or run along the shelf board.
- **SOP rule broken:** Steps 2.1 and 2.2. The cloth stays folded and flat, works the lip face only, and passes between grippers only by being laid on the cloth rest.
- **Coaching note:** folded and flat, down on the rest and up again, never across and never inside.

Violation: Wrong gripper dusted a half of the lip

- **Visible cue:** the **left gripper** carries the cloth past the shelf midline to the right. Or the **right gripper** carries it past the shelf midline to the left.
- **SOP rule broken:** Steps 2.1 and 2.2. The **left gripper** dusts the left half and the **right gripper** dusts the right half, each from the shelf midline outward.
- **Coaching note:** each gripper stops at the shelf midline. The cloth crosses only on the rest.

Violation: Lip not dusted as prescribed

- **Visible cue:** dust is still visible on the lip when the cloth goes back on its rest. A half gets one pass instead of two. The cloth lifts off the lip partway along a pass. Or a pass stops short of the end of the shelf.
- **SOP rule broken:** Steps 2.1 and 2.2. Keep the cloth pressed on the lip face. Run each half out to the end of the shelf and back, twice.
- **Coaching note:** on the lip face, out to the end, back, twice, and read it clean before the cloth goes down.

Violation: Dug under a stack

- **Visible cue:** a gripper moves, slides, or lifts a book that has another book lying on top of it. Or a book is pulled out from under a stack.

- **SOP rule broken:** Step 3.1. Work only free books, and let the book underneath become free on its own.
- **Coaching note:** never dig. The stacks are built with the taller book on top, so the free book is always the one you want.

Violation: Book taken out of height order

- **Visible cue:** the **left gripper** feeds a book that is visibly shorter as it lies than another free book still on the shelf. Or the finished row has a book taller than the one to its right.
- **SOP rule broken:** Steps 3.1 and 3.4. Take the longest free book each time, so the row runs tallest at the right end panel to shortest at the left.
- **Coaching note:** read all the free books before reaching in. Longest free book, every time.

Violation: Book fed wrong to the build place

- **Visible cue:** the book is pushed into the row or against the right end panel instead of stopping at the build place. It is pushed with an open gripper or by a wrist. It is left short of the build place. Or its spine is left not overhanging the lip.
- **SOP rule broken:** Step 3.2. Push the book with the closed **left gripper** as far as the build place, then push it forward until its spine overhangs the shelf lip.
- **Coaching note:** closed gripper, to the build place, then forward until a finger fits under the spine.

Violation: Book tipped up without the stop

- **Visible cue:** the **right gripper** starts the tip-up while the **left gripper** is away from the book's left end. The book skates left along the board instead of pivoting. Or the **left gripper** moves during the tip-up.
- **SOP rule broken:** Step 3.3. The **left gripper** stands hard against the book's left end as the stop and holds there until the book is standing.
- **Coaching note:** stop first, then tip. Without the block the book slides instead of standing.

Violation: Tip-up done wrong

- **Visible cue:** the **right gripper** closes on the pages, the page edge, or a corner instead of the covers on the overhanging spine strip. The pivot edge lifts off the board. The right gripper carries the book past leaning all the way upright. Or the book is rolled or turned end for end.
- **SOP rule broken:** Step 3.3. Close on the covers at the overhanging spine strip, keep the pivot edge on the board, and stop as soon as the book leans on the stop.
- **Coaching note:** covers only, pivot edge down, and hand the last of the lift to the left gripper.

Violation: Book shelved spine in, face out, or upside down

- **Visible cue:** a book in the row shows a cover or its page edge at the front instead of its spine. Or its spine title reads upside down.
- **SOP rule broken:** Steps 3.2 to 3.4. Feed the book spine forward with its bottom edge to the left, and tip it up to the left so it stands spine out and the right way up.
- **Coaching note:** check the lie before the tip. A book that lies wrong stands wrong.

Violation: Row not pressed home

- **Visible cue:** a gap shows between two books in the row. A book stands away from its neighbour. A gap shows between the tallest book and the right end panel. Or the **left gripper** never presses the book upright after the tip-up.
- **SOP rule broken:** Step 3.4. Press the book to the right with the **left gripper** until it comes upright and the row is snug against the right end panel.
- **Coaching note:** the tip-up only leans it. The press is what stands it up and closes the gap.

Violation: Left gripper grips or shoves instead of blocking and pressing

- **Visible cue:** the **left gripper** closes on a book. It pushes hard enough to bow a cover or skew the row. It drags a standing book sideways. Or it keeps pressing after the books have stopped moving.
- **SOP rule broken:** Steps 3.3 and 3.4. The stop and the steadying press are both a closed gripper held against the book, never a grip and never a shove.
- **Coaching note:** closed gripper, light pressure, stop as soon as the row stops moving.

Violation: Spine left sticking out past the lip

- **Visible cue:** a book in the finished row has its spine standing out past the shelf lip, or its spine sits visibly behind the spines beside it, and the **right gripper** never presses it back.
- **SOP rule broken:** Step 3.4. Press the new book's spine straight back with the closed **right gripper** until it sits just inside the shelf lip, level with the spines beside it.
- **Coaching note:** every book gets the back-press. That is what lines the fronts up.

Violation: Bookend set wrong

- **Visible cue:** the bookend lies on a face instead of standing on its base. Its flat faces end up facing front and back instead of left and right. It is turned on the way in. It is set on top of a book. Or it is set somewhere other than at the left end of the row.
- **SOP rule broken:** Steps 4.1 and 4.2. Carry the bookend in upright from the front with the **left gripper** and set it on its own base at the left end of the row, flat faces facing left and right.
- **Coaching note:** it goes in the same way it stood on its spot: on its base, faces left and right.

Violation: Bookend not pushed home, or pushed too far

- **Visible cue:** the bookend is left standing clear of the leftmost book. It is pushed while still gripped instead of pushed by the closed **left gripper**. It is pushed high on its face so it tips. Or the push carries on after the books stop and bows the row or shifts the bookcase.
- **SOP rule broken:** Step 4.2. Let go, come back low and closed on the bookend's left face, push until the row is snug, and stop there.
- **Coaching note:** low on the face, push until the books stop, then stop.

Violation: Bookend used as a support during shelving

- **Visible cue:** the bookend goes onto the shelf board before the sixth book is standing. Or it is used in place of the **left gripper's** stop or press while the row is still being built.
- **SOP rule broken:** Steps 3.3, 3.4, and 4.1. The row is held by the **left gripper** until all six books stand. The bookend goes in only after that.
- **Coaching note:** the left gripper holds the row. The bookend is the last thing in.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: dust left on a half of the lip, a book left leaning, a book out of height order, a spine left sticking out past the lip, or a non-book still on the board.
- **SOP rule broken:** Steps 1, 2.1, 2.2, 3.2, 3.3, 3.4, and 4.2. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, toppled, or knocked over

- **Visible cue:** a non-book or the bookend is dropped. A stack is knocked over. A book already standing is toppled or pushed off the shelf on the way in or out. The row collapses. Or the bin is knocked over.
- **SOP rule broken:** Steps 1 to 4. Nothing is dropped, toppled, or knocked out of its place, and every gripper leaves the shelf opening straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves. Draw straight out so a finger does not catch a spine.

Violation: Wrong arm used

- **Visible cue:** the **right gripper** touches a non-book, the bin, the bookend, or a stack. The **left gripper** tips a book up or reaches the right end panel. Or an object is handed from one gripper to the other in the air.
- **SOP rule broken:** Steps 1 to 4. The **right gripper** tips the books. The **left gripper** works the non-books, the feed, the stop, the press, and the bookend. Nothing changes hands in the air.

- **Coaching note:** right gripper tips, left gripper feeds and holds and finishes. The cloth only ever crosses on the rest.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a non-book on the shelf board, dust on the lip, a book still lying flat, fewer than six books standing, a book leaning, spine in, upside down, or out of height order, a gap in the row or at the right end panel, a spine sticking out past the lip, the bookend missing or clear of the row, the cloth off its rest, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5. Confirm the board, the row, the order, the snug fit, the fronts, the lip, the lower shelf, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the shelf lip or the book lengths**, so whether the lip is clean or whether the row is in height order cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Shelf opening too tight:** an opening so low that a book cannot tip up through it without touching the overhead shelf, or that a gripper cannot enter level from the front.
- **Bookcase not steady:** a unit that rocks, walks, or leans when a book is pushed along the board or the bookend is pushed home.
- **Right end panel loose:** a panel that shifts when the row is pressed against it.
- **Shelf board will not hold a book:** a board that is not level, or so slick that a correctly stood book will not stay where it is put, or so rough that a correct push will not slide a flat book along it.
- **Shelf lip too shallow to grip past:** a lip that will not let a book's spine overhang far enough for a gripper finger to pass under it.
- **Defective book:** one that will not stand on its own bottom edge unaided, that sags or flops when held correctly by its covers, or whose covers slip out of a correct grip.
- **Two books too close in length to tell apart**, so which free book is longest cannot be read from the front.

- **Defective bookend:** one that rocks on its base, will not stand upright unaided, is too light to hold six books, or slides along the board under a correct low push instead of moving the row.
- **Defective non-book:** one that comes apart, spills, or will not sit steady on the bare shelf board.
- **Defective cloth:** one that is torn, shedding, dry at episode start, or so loaded with dust that a correct pass only smears the lip.
- **Damaged lip surface:** a stain, scratch, or mark in the shelf lip itself that does not lift under a correct pass.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the bin, the bookend spot, the cloth rest, a stack, the build place, or the right end of the row cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Move one non-book from the shelf into the bin
2. Take the cloth from the cloth rest
3. Dust the left half of the shelf lip
4. Dust the right half of the shelf lip
5. Return the cloth to the cloth rest
6. Push one book along the board to the build place
7. Push the book forward so its spine overhangs the lip
8. Hold the stop against the book's left end
9. Tip one book up onto its bottom edge
10. Press the book upright into the row
11. Press the book's spine back off the lip
12. Set the bookend at the left end of the row
13. Push the bookend home so the row stands snug
14. Return both arms home and end the episode